Business Forecasting

Chapter 2:
Exploring Data Patterns and Choosing a Forecasting Technique

Course: BBAZ16605

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Thanks to Prof. Yan (Jenny) Chen

Recap …

- What is forecasting?
- How to forecast?
- Categories of forecasts
  - Qualitative vs. Quantitative
  - Time horizon
  - Micro vs. Macro
  - Nature of the output
- Selection of a forecasting method
- Forecasting steps

Three types of data

- One of the most time-consuming and difficult parts of forecasting is the collection of *valid and reliable data*.
  - "garbage in, garbage out"
- Information age!

Data is every where. But the difficult task facing most forecasters is how to find *relevant data* that will help solve their specific decision-making problems.

- Four criteria can be applied when determining whether data will be useful
  1. Data should be *reliable and accurate*.
  2. Data should be *relevant*.
  3. Data should be *consistent.*
  4. Data should be *timely*.

Three types of data

- *Cross-sectional* data are observations collected at a single point in time (all observations are from the same time period).
  - Example: age and maintenance cost.

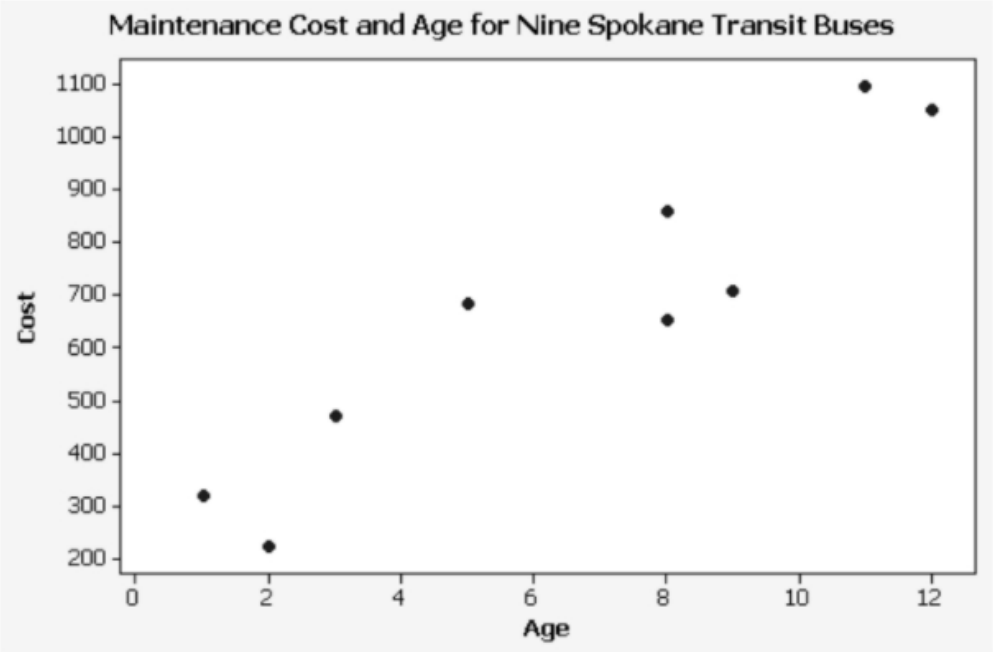


FIGURE 1 Scatter Diagram of Age and Maintenance Cost for Nine Spokane Transit Authority Buses

Three types of data

- *Cross-section data* is collected in a single time period and is described by individual units - people, houses, countries, etc. Some examples include:
- Student GPA at the end of this semester;
- Household expenses data of the last year - expenses on food, utility bills, income etc.
- Car data – engine power, color, price, speed.
- Ordering of the data does not matter!

Three types of data

- A *time series* consists of data that are collected, recorded or observed over successive increments of time.
  - Example: Monthly U.S. beer production

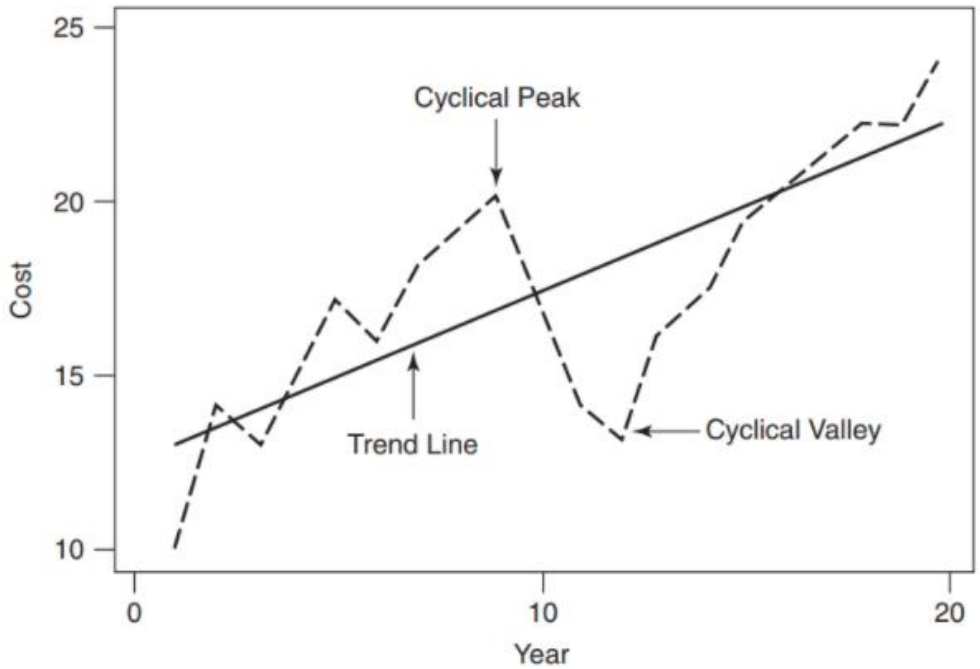


FIGURE 2 Trend and Cyclical Components of an Annual Time Series Such as Housing Costs

Three types of data

- Time series data refers to data captured at multiple points in time.
- Stock prices, interest rates, exchange rates, product prices, GDP, and so on are some examples.
- Time series data can be seen at various intervals, including hourly, daily, weekly, monthly, quarterly, and annual.
- In contrast to cross-sectional data, time-series data requires careful ordering.

  Each point depicts the values at various times in time. As a result, time series data are often presented in a logical order.
- Changing the order of the data ignores the data's temporal component!

Three types of data

- Longitudinal data (纵向数据)或 Panel data (面板数据)
  - Cross-sectional time series data
  - a (usually small) number of observations over time on a (usually large) number of cross-sectional units, like individuals, firms, or governments.

Three types of data

- Panel data is a combination of cross-sectional and time series data in which the same individuals (people, companies, cities) are observed at several points in time.

- A balanced panel exists when each individual has the same number of time period observations.

Longitudinal data / Panel data

Example [edit]

balanced panel:

person	year	income	age	sex
1	2001	1300	27	1
1	2002	1600	28	1
1	2003	2000	29	1
2	2001	2000	38	2
2	2002	2300	39	2
2	2003	2400	40	2

unbalanced panel:

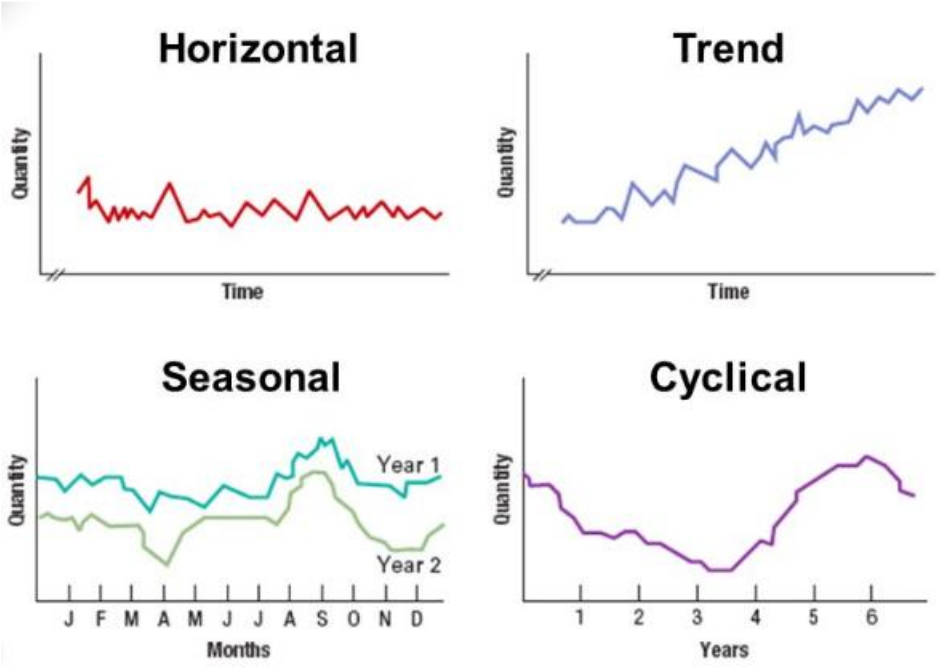
person	year	income	age	sex
1	2001	1600	23	1
1	2002	1500	24	1
2	2001	1900	41	2
2	2002	2000	42	2
2	2003	2100	43	2
3	2002	3300	34	1

Contain observations of multiple variables over multiple time periods for the same firms or individuals.

EXPLORING TIME SERIES DATA PATTERNS

Time Series Patterns

- Horizontal / Stationary in its mean
- Trend
- Seasonal
- Cyclic



Horizontal / Stationary in its mean

- Fluctuate around a constant level or mean
- Stationarity in its mean.
- a flat-looking series, without trend.

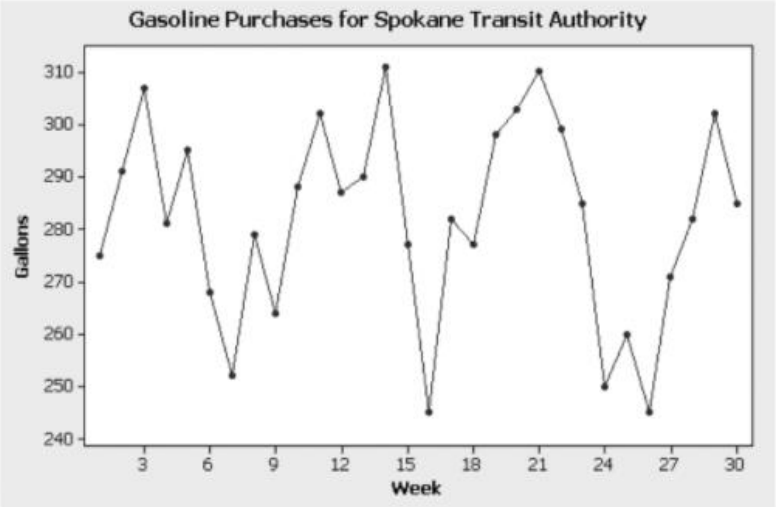


FIGURE 3 Time Series Plot of Weekly Gasoline Purchases for the Spokane Transit Authority for Example 2

Horizontal / Stationary in its mean

- Steady Daily Shuttle Times.
- Weekly/Monthly Grocery shopping.
- Number of classes/teaching load.

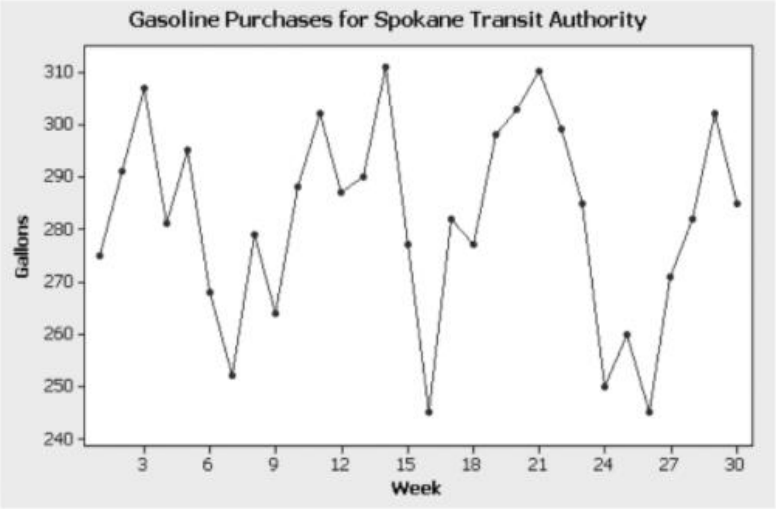
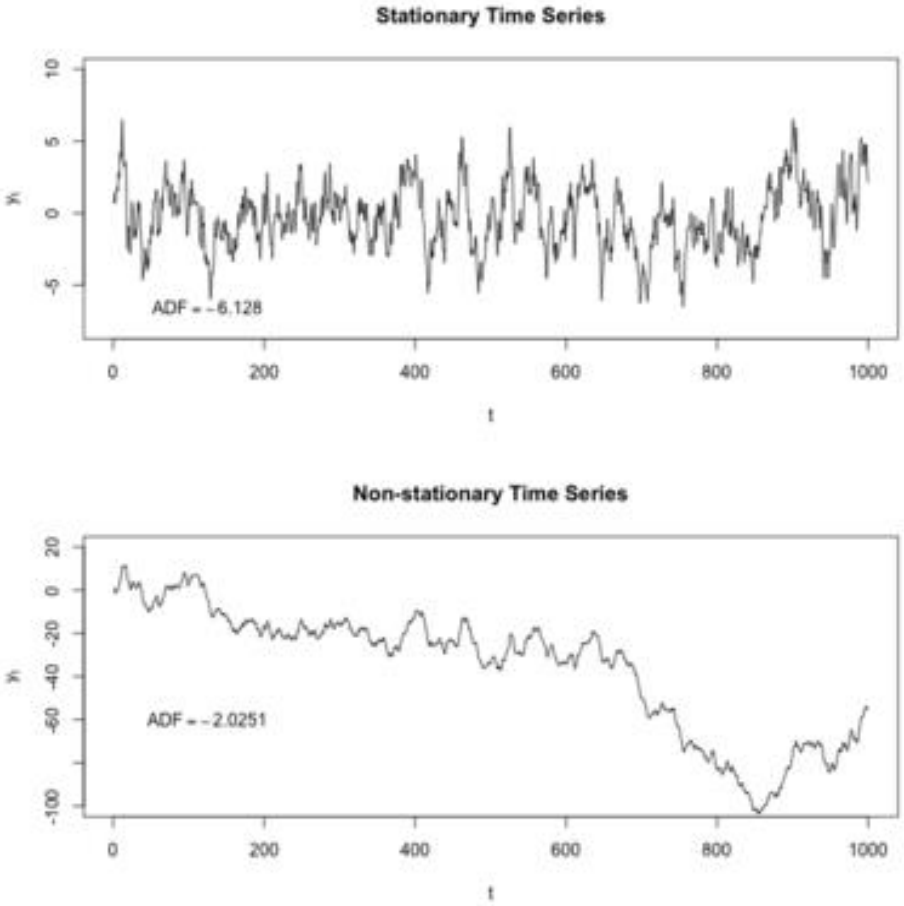


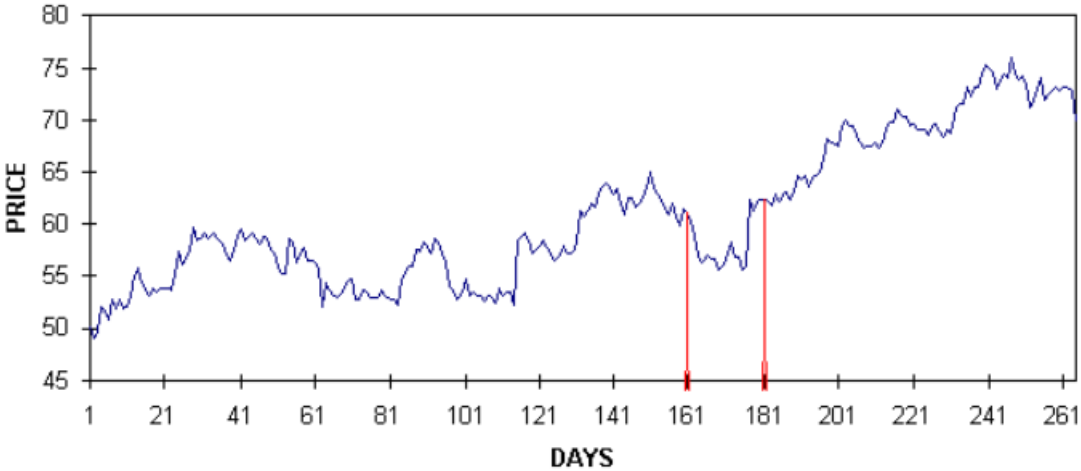
FIGURE 3 Time Series Plot of Weekly Gasoline Purchases for the Spokane Transit Authority for Example 2

Horizontal V.S. NonStationary



Trend

- Long-term growth or decline in the time series.
  - Macroeconomic variables
    - Gross national product
    - Employment
    - CPI (Consumer Price Index)
  - Basic forces behind trend-like behavior
    - Population growth
    - Productivity increase
    - Price inflation



Seasonal

- Fluctuation caused by seasonal factors.
  - Weather conditions, holidays, calendar
- Patterns of change repeats year after year.
  - Fixed cycle of 1 year
  - Monthly and quarterly

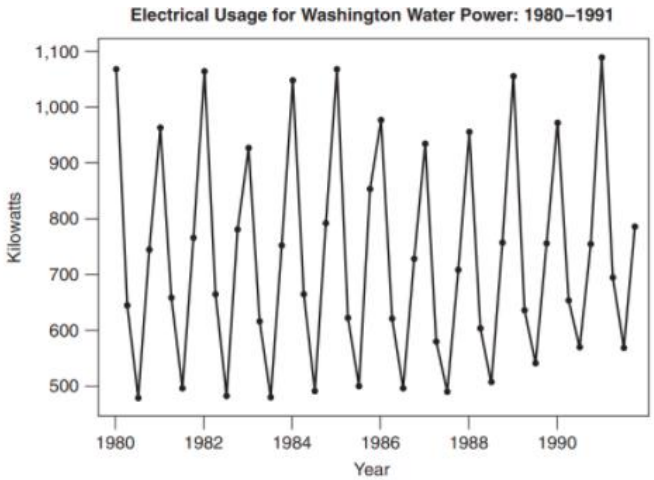


FIGURE 3 Electrical Usage for Washington Water Power Company, 1980–1991

Seasonal

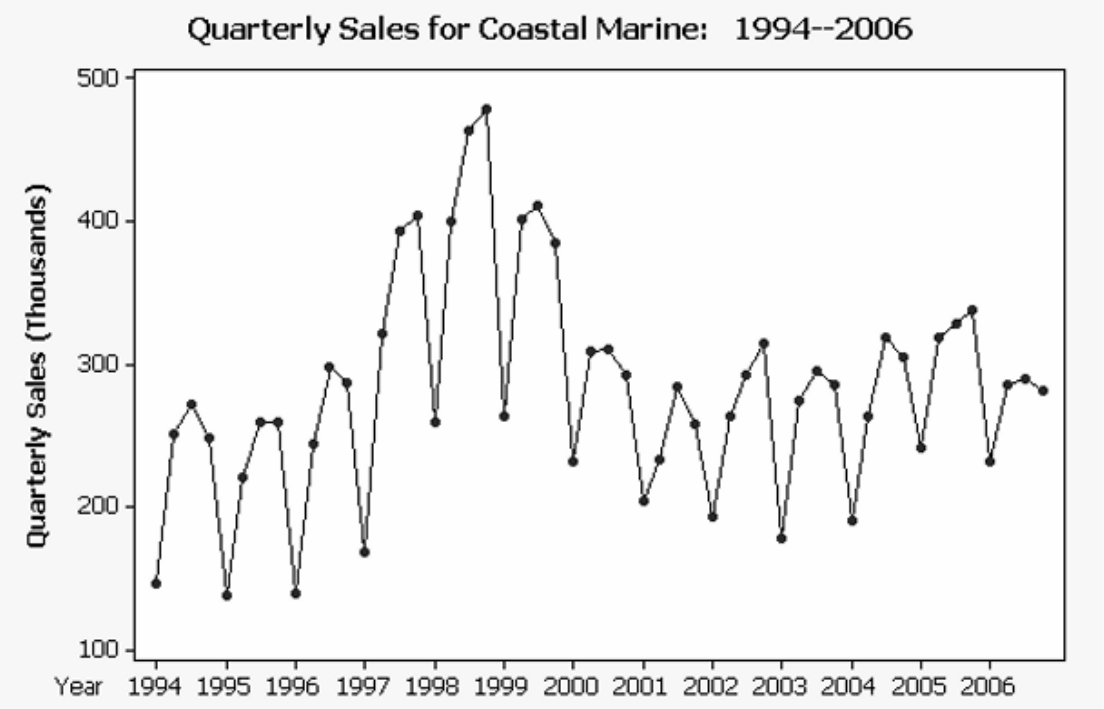


FIGURE 14 Time Series Plot of Quarterly Sales for Coastal Marine for Example 5

Cyclical

- Rises and falls are not of fixed period.
- Wavelike fluctuation around the long-term trend.
- One complete cycle is usually of at least 2 years.
  - Usually affected by business cycle
    - Economic expansions and contractions.
      - Unemployment rate
      - Investment
      - CPI
    - The Cyclical Nature of Fashion

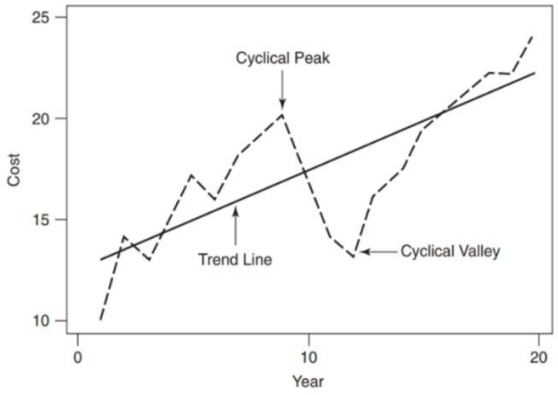


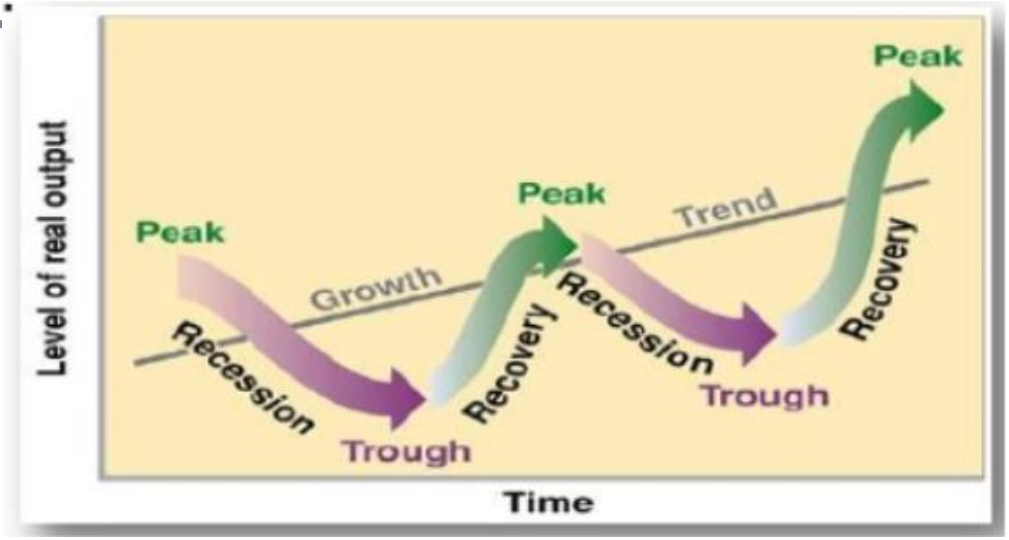
FIGURE 2 Trend and Cyclical Components of an Annual Time Series Such as Housing Costs

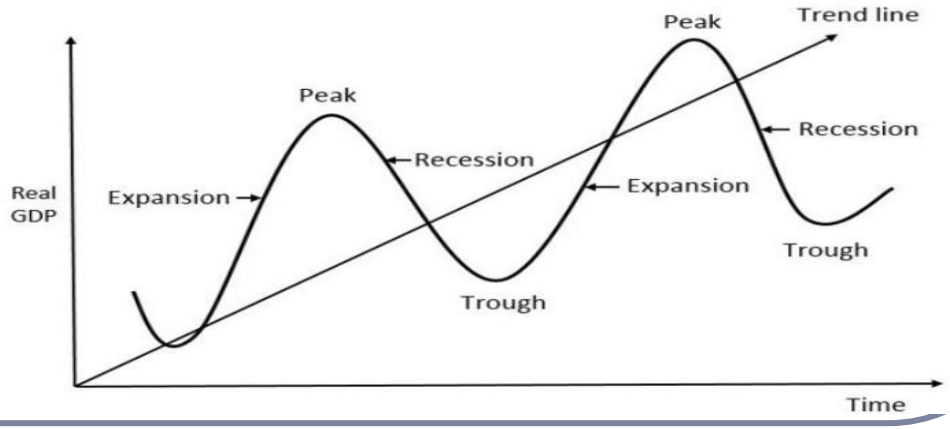
Business Cycle

4 Stages of the Business Cycle

- Expansion/Recovery
- Peak
- Contraction/Recession
- and valley/trough.

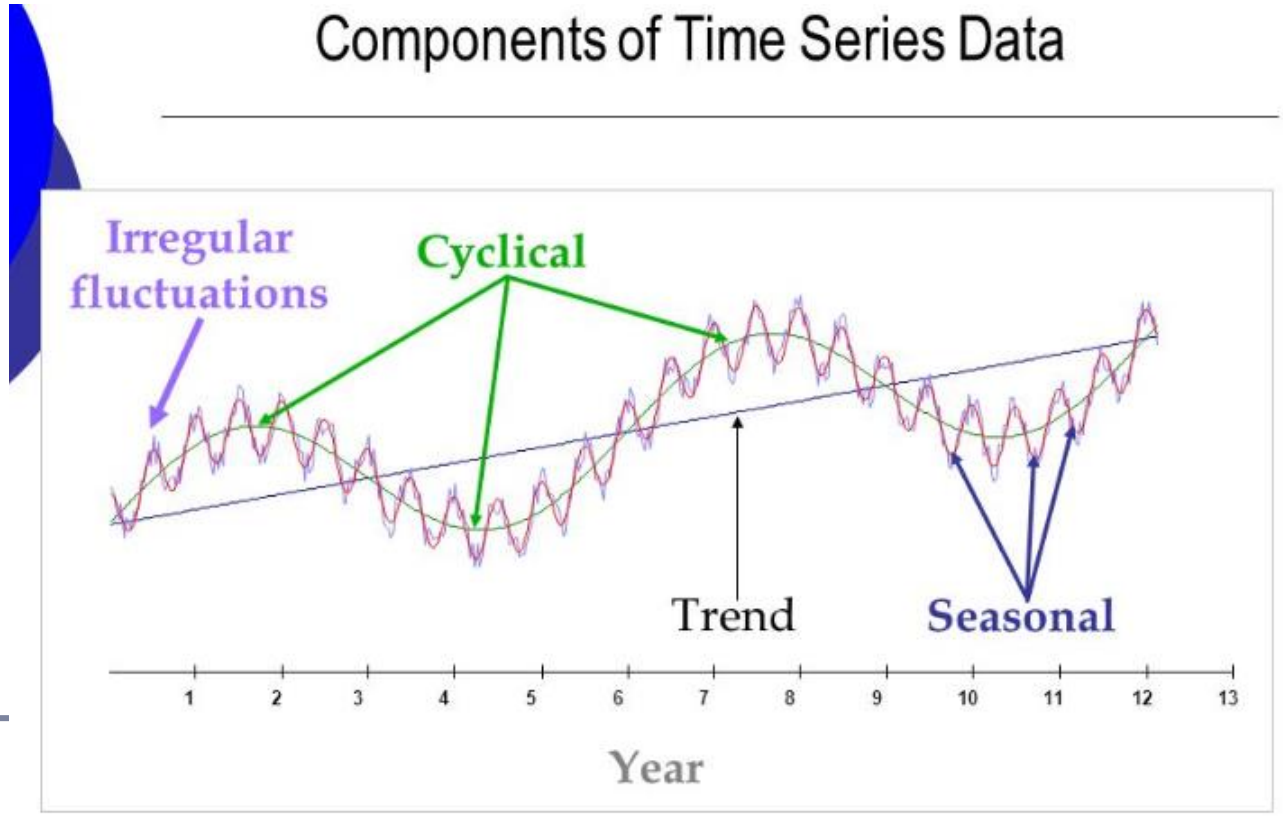
- —





Exploring Time Series Data Patterns

- Exhibits more than one pattern in a time series.



Decompose Time Series Data

- 4 components
  - Level component (水平分量)
    - 基准线，没有趋势，季节性和噪音时的基准水平
  - Trend component (趋势分量)
    - a long-term increase or decrease in the data
    - 增长或减少的部分
  - Seasonality (季节性分量)
    - regular fluctuations due to the season
    - 受季节性影响的部分
  - Irregular fluctuations / Random Variation (随机波動)
    - random variation/noise in the series.
    - 剩餘的或「包羅萬象」的因素，
    - 由那些影響時間數列的短期的、不可預期的和不重複出現的因素，因此無法預測

Exploring Data Patterns with Autocorrelation Analysis

- Correlation defines the statistical relationship between two variables. It measures the extent to which two variables are linearly related.

- *Autocorrelation* is the correlation between observations of the same variable at different time points.

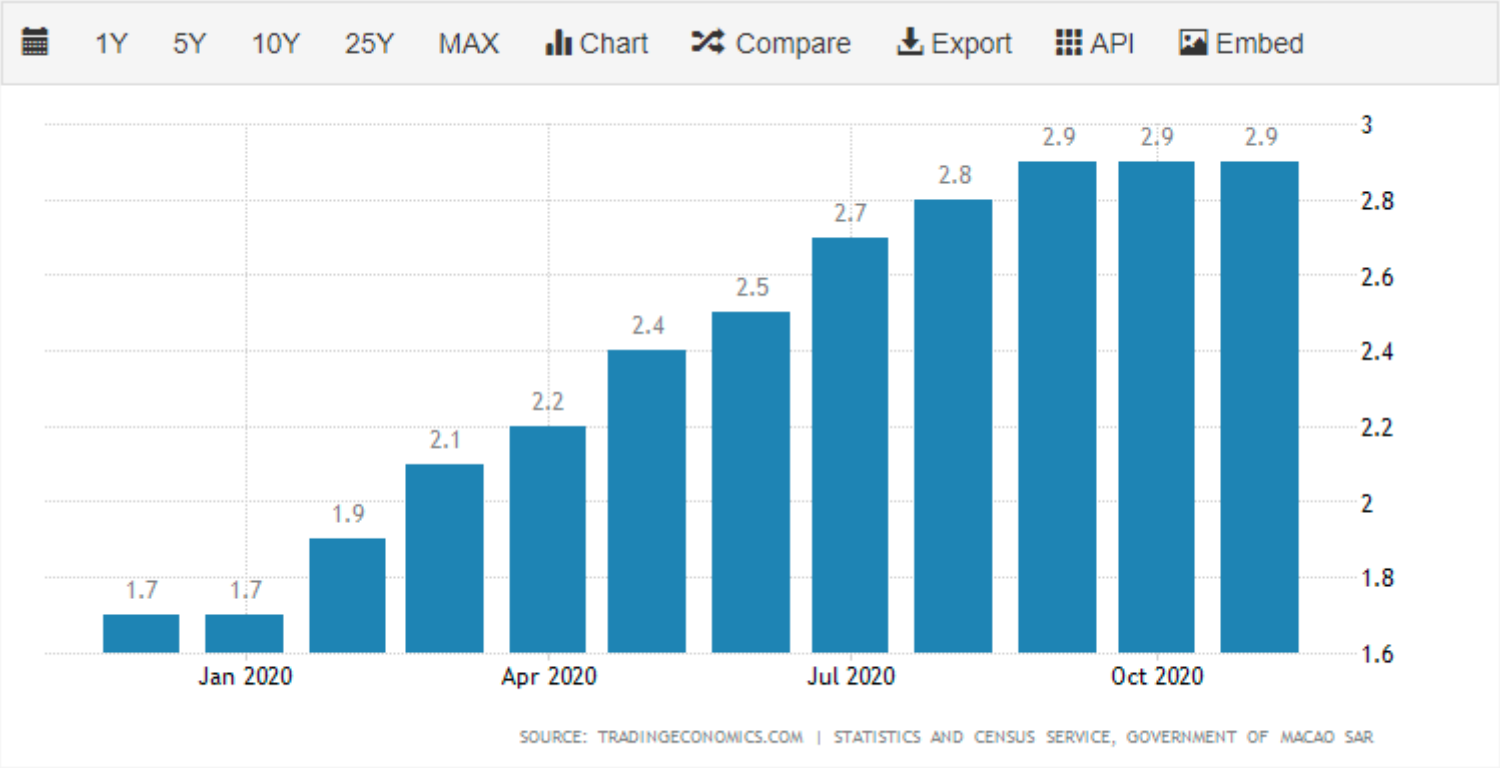
or

- Autocorrelation is the correlation between a variable lagged one or more time periods and itself.

- Autocorrelation is **the degree of similarity of the values** of the same variables over successive time intervals.

Exploring Data Patterns with Autocorrelation Analysis

- Data patterns, including components such as trend and seasonality, can be studied using autocorrelations.
- The patterns can be identified by examining the autocorrelation coefficients at different time lags of a variable.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Exploring Data Patterns with Autocorrelation Analysis

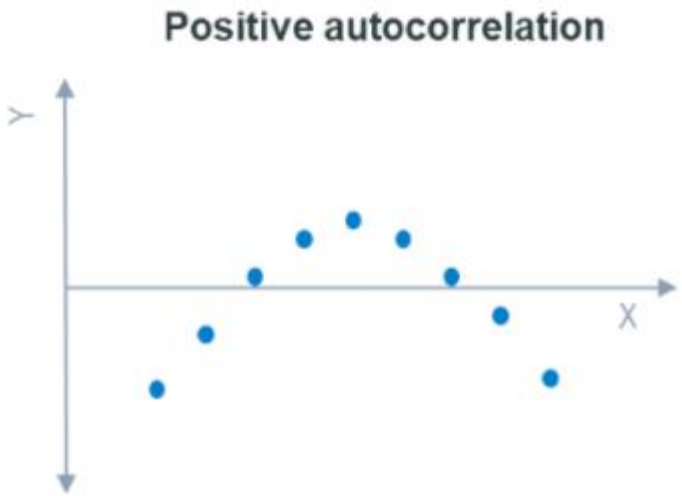
- Lag *k* autocorrelation coefficients

$$r_k = \frac{\sum_{t=k+1}^{n} (y_t - \bar{Y})(y_{t-k} - \bar{Y})}{\sum_{t=1}^{n} (y_t - \bar{Y})^2}$$

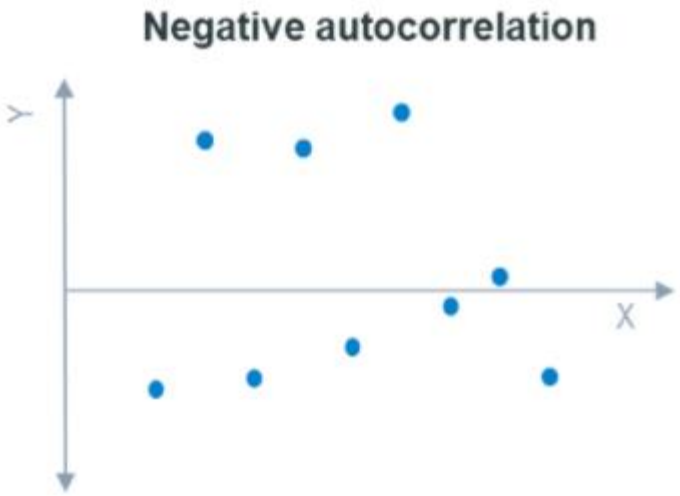
  - Measures similarity b/t observations of the same variable separated by *k* periods

- Data patterns can then be studied using autocorrelations.

Positive and negative autocorrelation



When data exhibiting positive **first-order correlation** is plotted, the points appear in **a smooth snake-like curve**,



With negative first-order correlation, the points form a **zigzag pattern** if connected

Exploring Data Patterns with Autocorrelation Analysis

- Example 1, P19

TABLE 1 VCR Data for Example 1

Time t	*Month*	*Original Data Y_t*	*Y Lagged One Period Y_{t-1}*	*Y Lagged Two Periods Y_{t-2}*
1	January	123		
2	February	130	123	
3	March	125	130	123
4	April	138	125	130
5	May	145	138	125
6	June	142	145	138
7	July	141	142	145
8	August	146	141	142
9	September	147	146	141
10	October	157	147	146
11	November	150	157	147
12	December	160	150	157

Equation 1 is the formula for computing the lag k autocorrelation coefficient (r_k) between observations, Y_t and Y_{t-k} that are k periods apart.

$$r_k = \frac{\sum_{t=k+1}^{n} (Y_t - \overline{Y})(Y_{t-k} - \overline{Y})}{\sum_{t=1}^{n} (Y_t - \overline{Y})^2} \qquad k = 0, 1, 2, \ldots \qquad \textbf{(1)}$$

where

r_k = the autocorrelation coefficient for a lag of k periods

$\overline{Y}$ = the mean of the values of the series

Y_t = the observation in time period t

Y_{t-k} = the observation k time periods earlier or at time period $t - k$

Exploring Data Patterns with Autocorrelation Analysis

- Lag *1* autocorrelation coefficients

TABLE 2 Computation of the Lag 1 Autocorrelation Coefficient for the Data in Table 1

Time t	Y_t	Y_{t-1}	$(Y_t - \bar{Y})$	$(Y_{t-1} - \bar{Y})$	$(Y - \bar{Y})^2$	$(Y_t - \bar{Y})(Y_{t-1} - \bar{Y})$
1	123	—	−19	—	361	—
2	130	123	−12	−19	144	228
3	125	130	−17	−12	289	204
4	138	125	−4	−17	16	68
5	145	138	3	−4	9	−12
6	142	145	0	3	0	0
7	141	142	−1	0	1	0
8	146	141	4	−1	16	−4
9	147	146	5	4	25	20
10	157	147	15	5	225	75
11	150	157	8	15	64	120
12	160	150	18	8	324	144
Total	1,704		0		1,474	843

$$\bar{Y} = \frac{1{,}704}{12} = 142$$

$$r_1 = \frac{843}{1{,}474} = .572$$

Exploring Data Patterns with Autocorrelation Analysis

- Lag *2* autocorrelation coefficients

$$r_2 = \frac{\sum_{t=2+1}^{n} (Y_t - \overline{Y})(Y_{t-2} - \overline{Y})}{\sum_{t=1}^{n} (Y_t - \overline{Y})^2} = \frac{682}{1{,}474} = .463$$

TABLE 1 VCR Data for Example 1

Time t	*Month*	*Original Data* Y_t	*Y Lagged One Period* Y_{t-1}	*Y Lagged Two Periods* Y_{t-2}
1	January	123		
2	February	130	123	
3	March	125	130	123
4	April	138	125	130
5	May	145	138	125
6	June	142	145	138
7	July	141	142	145
8	August	146	141	142
9	September	147	146	141
10	October	157	147	146
11	November	150	157	147
12	December	160	150	157